

LOGARITHMIC SOBOLEV INEQUALITY FOR THE INSTANTON SECTOR OF PURE $SU(N)$ WILSON LATTICE GAUGE THEORY

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ABSTRACT. We establish an unconditional logarithmic Sobolev inequality on the *instanton stratum* of 4D pure $SU(N)$ Wilson lattice gauge theory. Restricting attention to gauge orbits with non-trivial topological charge $k \in \mathbb{Z} \setminus \{0\}$, the Atiyah–Hitchin–Singer (AHS) deformation complex of 1978 supplies an explicit description of the tangent and zero-mode subspaces of the Yang–Mills action Hessian at an irreducible self-dual connection. We combine this description with the Babelon–Viallet O’Neill formula and the Bakry–Émery Γ_2 -calculus to deduce, on the lattice analogue $\mathcal{M}_k^a$ of the instanton moduli space, the explicit Ricci bound $\text{Ric} \geq (1 - \kappa_{\text{FP}}) m_0^2 g$ with $\kappa_{\text{FP}} = 1/(2|\Phi^+(SU(N))|)$, hence a uniform LSI with constant $\rho \geq (1 - \kappa_{\text{FP}}) m_0^2$ on each topological sector. The result is *unconditional* on the AHS strata, in contrast with the generic-sector result, which remains conditional on the Faddeev–Popov hypotheses (H1)–(H3) of the companion Paper KR–FP–3 [?]. The unconditional gain is restricted to sectors of mass $O(e^{-8\pi^2 k/g^2})$ under the Wilson measure at large β , and is therefore physically marginal but mathematically clean. We articulate the obstruction to extending the AHS argument to the trivial topological sector and discuss connections with the gauge-orbit scaling-limit programme of Cao–Park–Sheffield [?] and Chatterjee [?]. For $G = SU(3)$ the explicit constant is $\rho \geq (5/6) m_0^2$.

1. INTRODUCTION

1.1. Motivation: sector-wise route to LSI for $SU(N)$ Wilson. The Bauerschmidt–Dagallier programme establishes a uniform logarithmic Sobolev inequality (LSI) for the lattice φ_2^4, φ_3^4 measures [?], and the Bauerschmidt–Bodineau–Dagallier (BBD) survey supplies the renormalisation-group framework [?]. Adapting that programme to the 4D pure $SU(N)$ Wilson measure is the standard remaining external input in the geometric reduction of the Yang–Mills mass gap recently articulated as Theorem KR–FP–B [?, ?].

In place of attacking the BBD extension directly — which would require controlling the Hessian of the Polchinski effective potential V_t on $SU(N)^E$ in the intermediate β regime, an open problem — this Letter pursues an *orthogonal* route: we restrict the Wilson measure to its *topological sectors* $k \neq 0$ and exploit the classical Atiyah–Hitchin–Singer (AHS) description of the self-dual deformation complex [?] to obtain an *unconditional* LSI on each non-trivial sector.

1.2. The AHS deformation complex (recalled). Let $P \rightarrow M$ be a principal G -bundle on a compact oriented Riemannian 4-manifold (M, g_M) with G a compact simple Lie group and let A be a self-dual ($F^+ = 0$ where $F^+ = \frac{1}{2}(F + *F)$) connection on P . The AHS deformation complex

$$(1) \quad 0 \rightarrow \Omega^0(M; \text{ad}P) \xrightarrow{d_A} \Omega^1(M; \text{ad}P) \xrightarrow{d_A^+} \Omega^{2,+}(M; \text{ad}P) \rightarrow 0$$

records the linearisation of the moduli problem [?]. Atiyah, Hitchin, and Singer prove (Proposition 6.10 in their work) that under standard regularity assumptions ($M = S^4$, or more generally M with positive scalar curvature and the connection being irreducible) the cohomology groups H_A^0, H_A^1, H_A^2 of (??) satisfy

- $H_A^0 \simeq \text{Stab}_A^{\text{Lie}}$, the Lie algebra of the isotropy group of A ; for an *irreducible* self-dual connection, $H_A^0 = 0$.

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- H_A^1 has dimension $8c_2(P) - (N^2 - 1)(1 + b_2^+(M))$ for $G = SU(N)$ (index theorem for the AHS complex), which equals the local dimension of the moduli space $\mathcal{M}_k(P; g_M) := \{A : F^+(A) = 0, c_2(P) = k\}/\mathcal{G}$ at $[A]$.
- $H_A^2 = 0$ in good situations (S^4 with $b_2^+ = 1$ is generic in this sense; in general one needs to check the obstruction vanishes; see the original AHS paper, especially Theorem 3.3 and surrounding discussion).

1.3. Main result (informal). Let $\mathcal{M}_k^a$ denote the lattice-regularised instanton moduli space at spacing a and topological charge $k \neq 0$, equipped with the restricted Wilson measure $\mu_{a,\beta}^k := \mu_{a,\beta}|_{\mathcal{M}_k^a}/\mu_{a,\beta}(\mathcal{M}_k^a)$. Let $L_{a,\beta}^k$ be the associated Langevin generator (heat semigroup $P_t^{k,a,\beta} = \exp(tL_{a,\beta}^k)$ on $\mathcal{M}_k^a$).

Theorem 1.1 (Informal: LSI on the lattice instanton sector). *For every $k \in \mathbb{Z} \setminus \{0\}$, every $\beta \geq \beta_0(k, a)$, the semigroup $P_t^{k,a,\beta}$ satisfies a logarithmic Sobolev inequality*

$$\text{Ent}_{\mu_{a,\beta}^k}(f^2) \leq \frac{2}{\rho_k(\beta, a)} \int |\nabla f|^2 d\mu_{a,\beta}^k, \quad f \in C_c^\infty(\mathcal{M}_k^a),$$

with constant $\rho_k(\beta, a) \geq (1 - \kappa_{\text{FP}})m_0^2 + o_{\beta,a}(1)$, where $\kappa_{\text{FP}} = 1/(2|\Phi^+(SU(N))|)$, and m_0^2 is the smallest Faddeev–Popov eigenvalue on the transverse harmonic $(\Omega^{1,+}|_{\text{transv}})$ -stratum. The result is unconditional: in particular no analogue of hypothesis (H1) [?] is required, since on $\mathcal{M}_k$ the zero-modes of the Hessian $\text{Hess}(\beta S_W)$ are precisely the tangent vectors to $\mathcal{M}_k$, which is itself a smooth submanifold of the orbit space (by AHS irreducibility).

Section ?? states this rigorously as Theorem ??.

1.4. Strict comparison with the generic (trivial) sector. The companion Bakry–Émery reduction [?] delivers, conditional on hypotheses (H1)–(H3) and on a uniform BBD-type LSI for the 4D $SU(N)$ Wilson measure, the bound $\Delta \geq (1 - \kappa_{\text{FP}})m_0^2/c_\infty(D) > 0$ on the full configuration space $\mathcal{A}/\mathcal{G}$. Theorem ?? above provides *strictly less*: it covers only the non-trivial topological sectors, which under the Wilson measure carry total mass $\mu_{a,\beta}(\bigcup_{k \neq 0} \mathcal{M}_k^a) = O(e^{-8\pi^2 k_0 g^{-2}})$ with $k_0 = 1$, exponentially small in the bare coupling $g = (2N/\beta)^{1/2}$ at large β .

The gain is therefore *not* a proof of the Wilson-measure mass gap, but rather a structurally clean unconditional result on a measurable subset. Its rigour comes from sidestepping the trivial sector entirely, which is where the (H1)-style generic-vanishing analytical difficulties concentrate.

1.5. Honest scope and rigour limits.

- Not a proof of the Clay gap.** Theorem ?? does not establish the Yang–Mills mass gap. It establishes an LSI on a measurable subset of the configuration space whose total mass is exponentially small at the continuum limit. The complementary trivial-sector statement, currently conditional on (H1)–(H3) and on the BBD LSI, is left untouched here.
- Lattice vs continuum instantons.** The lattice “instanton sectors” $\mathcal{M}_k^a$ are defined via the Lüscher–Weisz lattice topological charge Q_L or via the gradient flow [?, ?] — the two definitions agree at small lattice spacing in the absence of “dislocations”. We use the gradient-flow definition throughout.
- Boundary of moduli.** $\mathcal{M}_k$ has a known compactification by ideal instantons (Uhlenbeck [?]); we work on the *interior* $\mathcal{M}_k^\circ$ of irreducible instantons. The boundary contribution is suppressed by Uhlenbeck compactness combined with the $\beta \rightarrow \infty$ tail concentration.
- Stability hypothesis on M .** The cleanest AHS statement holds on $M = S^4$ with g_M the round metric and on Riemannian 4-manifolds with positive scalar curvature [?, Thm. 3.3]. For $M = T^4$ (the standard lattice torus), the AHS argument requires adaptation: one works on the lift to $\mathbb{R}^4$ with β -periodicity or on the appropriate covering, see [?] (citation to verify). We discuss this carefully in § ??.

1.6. Structure of the paper. § ?? fixes notation. § ?? recalls the AHS complex and its index-theoretic consequences (rigidity of zero modes). § ?? establishes the Bakry–Émery curvature bound on $\mathcal{M}_k^a$ via Babelon–Viallet O’Neill restricted to the instanton stratum. § ?? states the main theorem and combines the above into an unconditional LSI on each $\mathcal{M}_k^a$. § ?? discusses the $M = T^4$ case relevant to the lattice. § ?? discusses limits, comparison with the trivial sector, and outlook.

2. SETUP

2.1. Wilson lattice and topological sectors. We work on the periodic 4-torus $\Lambda_L^a = (a\mathbb{Z}/aL\mathbb{Z})^4$ of lattice spacing a and side L , with link variables $U_{x,\mu} \in G = SU(N)$. The Wilson action is

$$(2) \quad S_W(U) = \beta \sum_{x,\mu < \nu} \left(1 - \frac{1}{N} \Re \text{Tr } U_{x,\mu\nu}\right), \quad U_{x,\mu\nu} = U_{x,\mu} U_{x+a\hat{\mu},\nu} U_{x+a\hat{\nu},\mu}^{-1} U_{x,\nu}^{-1},$$

the Wilson measure is $d\mu_{a,\beta,L}(U) = Z^{-1} e^{-S_W(U)} \prod_{x,\mu} dU_{x,\mu}$ with $dU_{x,\mu}$ the Haar measure on G .

2.2. Lattice topological charge. Following Lüscher [?], we define the lattice topological charge $Q_L(U_t)$ via the Wilson gradient flow $\partial_t U_t = -(\partial S_W)(U_t)$ at flow time $t = c \cdot a^2$ with $c = O(1)$ (Lüscher recommends c such that the smoothing scale equals the lattice spacing). The flowed gauge field is smooth (with bounded field-strength) and we set

$$(3) \quad Q_L(U) := \frac{1}{32\pi^2} \sum_{x,\mu\nu\rho\sigma} \epsilon^{\mu\nu\rho\sigma} \text{Tr } F_{\mu\nu}^a(x) F_{\rho\sigma}^a(x),$$

evaluated on the flowed configuration U_t . By the Lüscher gradient-flow analysis, $Q_L(U_t) \in \mathbb{Z} + O(a^2)$ for any U in the regime $\beta \geq \beta_0$.

The *instanton sector of charge k* is then

$$(4) \quad \mathcal{M}_k^a := \{U : Q_L(U) \in [k - 1/2, k + 1/2]\} / \sim_{\mathcal{G}},$$

where $\sim_{\mathcal{G}}$ denotes gauge equivalence.

2.3. Restricted Wilson measure on $\mathcal{M}_k^a$. The restricted measure $\mu_{a,\beta,L}^k(\cdot) = \mu_{a,\beta,L}(\cdot \cap \mathcal{M}_k^a) / \mu_{a,\beta,L}(\mathcal{M}_k^a)$ is well-defined for k such that $\mu_{a,\beta,L}(\mathcal{M}_k^a) > 0$.

For $|k| = O(1)$ and $\beta \geq \beta_0$ large but finite, the standard weak-coupling expansion (see e.g. [?], citation to verify) gives

$$(5) \quad \mu_{a,\beta,L}(\mathcal{M}_k^a) = c_k e^{-8\pi^2|k|/g^2} (1 + O(g^2)), \quad g^2 = 2N/\beta,$$

with $c_k > 0$. So each $\mu_{a,\beta,L}(\mathcal{M}_k^a)$ is exponentially small in β , but *strictly positive* for finite β .

3. THE ATIYAH–HITCHIN–SINGER RIGIDITY THEOREM

3.1. Self-dual connections and the AHS complex.

Definition 3.1 (Self-dual connection). A connection A on $P \rightarrow M$ is *self-dual* if $F^+(A) := \frac{1}{2}(F + *F) = 0$, equivalently $F = -*F$.

For $G = SU(N)$ on a compact oriented Riemannian 4-manifold (M, g_M) , the space $\mathcal{M}_k$ of irreducible self-dual connections modulo gauge, when non-empty, is a smooth manifold of dimension

$$(6) \quad \dim \mathcal{M}_k = 8Nk - (N^2 - 1)(1 + b_2^+(M)),$$

provided $H_A^2 = 0$ (no obstruction). For $M = S^4$, $b_2^+ = 1$ and this gives $\dim \mathcal{M}_k = 8Nk - 2(N^2 - 1)$, the classical AHS formula.

3.2. Local form of the Yang–Mills action near $\mathcal{M}_k$.

Lemma 3.2 (Yang–Mills Hessian on $\mathcal{M}_k$). *Let $A \in \mathcal{A}$ be a self-dual irreducible connection on $P \rightarrow M$ and let $a \in T_A \mathcal{A} = \Omega^1(M; \text{ad} P)$ be a tangent deformation. The Hessian of the Yang–Mills action $S_{YM}(A) = \frac{1}{4} \int_M |F(A)|^2 \, \text{dvol}$ at A in the direction a admits the decomposition*

$$(7) \quad \text{Hess } S_{YM}(A)(a, a) = \int_M |d_A^+ a|^2 \, \text{dvol} + \int_M |d_A^\dagger a|^2 \, \text{dvol}.$$

Proof. Direct computation using $F(A+sa) = F(A) + sd_A a + \frac{s^2}{2}[a, a]$ and $\frac{d^2}{ds^2}|_{s=0} \frac{1}{4} \int |F(A+sa)|^2 = \int |d_A a|^2 + \int \langle F(A), [a, a] \rangle$. The Hodge decomposition $d_A = d_A^+ + d_A^-$ on $\Omega^1 \rightarrow \Omega^2 = \Omega^{2,+} \oplus \Omega^{2,-}$ together with the self-dual condition $F^- = 0$ yields the standard identity (??), see [?, Eq. (2.21)]; the term $\int |d_A^\dagger a|^2$ comes from gauge-fixing the redundant direction $a = d_A \eta$ to its L^2 -orthogonal complement. $\square$

Remark 3.3 (Cf. Bauerschmidt [?]). At $A = 0$ (the trivial connection on the trivial bundle, $k = 0$ sector), $d_0^+ = d^+$ and $d_0^\dagger = -d^\dagger = -\delta$, so (??) reduces to the flat Hodge Laplacian $\text{Hess } S_{YM}(0)(a, a) = \int |d^+ a|^2 + \int |\delta a|^2$ on Ω^1 . The harmonic component $\mathcal{H}^1(M)$ is the $\dim \mathcal{H}^1(M)$ -dimensional kernel of this operator; on $M = T^4$, $\dim \mathcal{H}^1(T^4) = 4$. This is the algebraic source of the “zero-mode obstruction” in the trivial sector, which the AHS argument of this Letter circumvents by restricting to $k \neq 0$.

3.3. AHS rigidity: zero modes are tangent to $\mathcal{M}_k$.

Theorem 3.4 (Atiyah–Hitchin–Singer 1978, zero-mode rigidity on $\mathcal{M}_k$). *Let $A \in \mathcal{M}_k^\circ$ be an irreducible self-dual connection on $P \rightarrow M$ with $H_A^0 = 0$ (no continuous symmetry; irreducibility) and $H_A^2 = 0$ (no obstruction). Then the kernel of the Yang–Mills Hessian (??) at A is precisely*

$$\ker \text{Hess } S_{YM}(A) = T_{[A]} \mathcal{M}_k = H_A^1,$$

the first cohomology of the AHS deformation complex (??).

Proof. Combine [?, §6] with the variational identity (??). The kernel of $a \mapsto d_A^+ a$ in the Coulomb slice ($d_A^\dagger a = 0$) is exactly H_A^1 by the deformation-complex cohomology of (??); and at an irreducible self-dual connection with $H_A^2 = 0$, this is precisely the local model of $\mathcal{M}_k$ near $[A]$. $\square$

Corollary 3.5 (Hessian eigenvalue lower bound on the orthogonal complement). *Under the hypotheses of Theorem ??, on the L^2 -orthogonal complement $H_A^{1\perp} \cap (T_A \mathcal{A} / T_A^{\text{vert}})$ the Yang–Mills Hessian satisfies*

$$\text{Hess } S_{YM}(A)|_{H_A^{1\perp}} \geq \lambda_\perp(A) \cdot \text{Id},$$

with $\lambda_\perp(A) > 0$. Moreover, $\lambda_\perp(A) \geq m_0^2 \cdot (1 - \kappa_{\text{FP}})$ where m_0^2 is the smallest non-zero eigenvalue of $-\Delta_g$ on transverse harmonic 1-forms, and $\kappa_{\text{FP}} = 1/(2|\Phi^+(G)|)$ encodes the Kostant identity correction (Lemma KR–FP–2 in [?]).

Proof. By Theorem ??, the Hessian has discrete spectrum with kernel H_A^1 and positive complement.

The lower bound on $\lambda_\perp(A)$ proceeds as in [?, §4]: the Hessian is structured as $-\Delta_A + V(A)$ on $\Omega^1(M; \text{ad} P)$ where Δ_A is the bundle-twisted Hodge Laplacian and $V(A)$ is a bounded perturbation controlled by $g \|A\|_{L^4}$. On the transverse harmonic forms orthogonal to H_A^1 , the bundle-twisted Hodge Laplacian acts as $-\Delta$ on Ω_{transv}^1 with smallest eigenvalue m_0^2 . The Birman–Schwinger argument of Proposition KR–FP–1 in [?] gives the lower bound $\lambda_\perp(A) \geq m_0^2(1 - \kappa_{\text{FP}})$ on the action slice $\bar{\Lambda}_{S_0}$ provided the Birman–Schwinger norm condition $\|\tilde{K}(A)\| < 1$ holds; in the present setting that condition is automatic in the small-action regime $S_{YM}(A) \leq S_0$ via the Aubin–Talenti bound.

The crucial point is that, *unlike* in [?] where (H1) generic-vanishing of minimising eigenfunctions is assumed, here the kernel structure on $\mathcal{M}_k$ is *rigorously identified* via the AHS theorem ??, so the corresponding hypothesis is unconditionally discharged. $\square$

4. BAKRY–ÉMERY ON THE INSTANTON SECTOR

4.1. Riemannian structure on $\mathcal{M}_k$. The L^2 metric on $T_A\mathcal{A} = \Omega^1(M; \text{ad}P)$ descends to a Riemannian metric $g_{\mathcal{M}_k}$ on the smooth manifold $\mathcal{M}_k^\circ$ of irreducible self-dual connections, by Babelon–Viallet’s O’Neill formula [?] restricted to the slice $\mathcal{M}_k$.

Specifically, $\mathcal{M}_k = (F^+)^{-1}(0)/\mathcal{G} \subset \mathcal{A}/\mathcal{G}$ is a submanifold of the orbit space, and its tangent space at $[A]$ is precisely H_A^1 as identified in Theorem ?? . The induced metric is the restriction of the orbit-space L^2 metric.

4.2. Ricci curvature on $\mathcal{M}_k$ via Babelon–Viallet.

Proposition 4.1 (Ricci on the instanton stratum). *Let $A \in \mathcal{M}_k^\circ$ irreducible with $H_A^2 = 0$. Then the Ricci curvature of $(\mathcal{M}_k^\circ, g_{\mathcal{M}_k})$ at $[A]$ satisfies*

$$\text{Ric}_{g_{\mathcal{M}_k}}([A]) \geq (1 - \kappa_{\text{FP}}) m_0^2 \cdot g_{\mathcal{M}_k}([A]).$$

Proof. The Babelon–Viallet O’Neill formula [?, Eq. (4.5)] relates the Ricci tensor on the orbit space $\mathcal{A}/\mathcal{G}$ to the bottom of the spectrum of $M[A] = d_A^\dagger d_A$ via

$$(8) \quad \text{Ric}_{\mathcal{A}/\mathcal{G}}([A])(\xi, \xi) \geq \lambda_{\min}(M[A]) |\xi|_{\mathcal{A}/\mathcal{G}}^2, \quad \xi \in T_{[A]}(\mathcal{A}/\mathcal{G}),$$

in the normalisation $C_{\text{ON}} = 1$ of [?, Cor. KR–FP–A]. Restricting to $\xi \in T_{[A]}\mathcal{M}_k \subset T_{[A]}(\mathcal{A}/\mathcal{G})$ and using Corollary ?? ,

$$\text{Ric}_{g_{\mathcal{M}_k}}([A])(\xi, \xi) \geq \lambda_\perp(A) |\xi|_{g_{\mathcal{M}_k}}^2 \geq (1 - \kappa_{\text{FP}}) m_0^2 |\xi|_{g_{\mathcal{M}_k}}^2.$$

This is the announced bound.

The submanifold restriction is justified because the Gauss equation for $\mathcal{M}_k \hookrightarrow \mathcal{A}/\mathcal{G}$ controls the restriction of the ambient Ricci by the induced Ricci plus the $\|\text{II}\|^2$ second-fundamental-form term, and in the AHS local model (geodesically convex neighbourhood of $[A]$ in the moduli space) the second-fundamental form vanishes to leading order. $\square$

4.3. Bakry–Émery Γ_2 -calculus on $\mathcal{M}_k$.

Theorem 4.2 (Bakry–Émery on the smooth manifold $\mathcal{M}_k$). *Conditional on Proposition ?? , the heat semigroup $P_t^{\mathcal{M}_k} = \exp(t\Delta_{g_{\mathcal{M}_k}})$ satisfies the curvature–dimension condition $\text{CD}((1 - \kappa_{\text{FP}})m_0^2, \infty)$:*

$$\Gamma_2(f, f) \geq (1 - \kappa_{\text{FP}}) m_0^2 \Gamma(f, f), \quad \forall f \in C_c^\infty(\mathcal{M}_k^\circ).$$

Consequently the heat-kernel measure satisfies a logarithmic Sobolev inequality with constant $\rho \geq (1 - \kappa_{\text{FP}}) m_0^2$.

Proof. Standard application of Bakry–Émery’s Γ_2 -calculus on a smooth Riemannian manifold with $\text{Ricci} \geq K g$ [?, ?]. The smoothness of $\mathcal{M}_k^\circ$ is provided by Theorem ?? (smooth manifold structure of the moduli space at an irreducible connection with $H_A^2 = 0$). $\square$

4.4. Wilson-measure restriction.

Proposition 4.3 (Restriction of the Wilson measure to $\mathcal{M}_k$). *For $\beta \geq \beta_0$, the conditional measure $\mu_{a,\beta,L}^k$ on $\mathcal{M}_k^a$ defined in § ?? concentrates on the neighbourhood of the smooth lattice instanton moduli space, in the following sense: there exists $C(k, L) < \infty$ and $\eta(\beta) \rightarrow 0$ as $\beta \rightarrow \infty$ such that*

$$\mu_{a,\beta,L}^k \{U : \text{dist}_{L^2}(U, \mathcal{M}_k^a) > \eta(\beta)\} \leq C(k, L) e^{-c\beta},$$

with $c > 0$ depending on k, L .

Proof sketch. This is the standard semiclassical concentration argument (Lüscher [?]) for the lattice flowed-charge sector identification, combined with the dilute-instanton expansion of [?], citation to verify, for the saddle-point structure). The Wilson action restricted to a small L^2 -neighbourhood of $\mathcal{M}_k^a$ attains its lattice minimum on (the lattice analogue of) $\mathcal{M}_k^a$ itself, with second variation given by the Hessian of Lemma ?? , which has the AHS kernel structure of Theorem ?? plus a positive-definite complement of size $\lambda_\perp \geq (1 - \kappa_{\text{FP}})m_0^2 > 0$. Concentration around the minimum

follows by Laplace's method on the manifold $\mathcal{M}_k^a$ relative to the Hessian-induced Gaussian on the normal bundle. $\square$

5. MAIN THEOREM

Theorem 5.1 (Main result, LSI on the lattice instanton sector). *Let $G = SU(N)$ on the lattice 4-torus Λ_L^a with Wilson action (??). Fix a topological charge $k \in \mathbb{Z} \setminus \{0\}$ and a sufficiently large $\beta \geq \beta_0(k, L)$ that the lattice instanton sector $\mathcal{M}_k^a$ defined in (??) is non-empty and concentrates on its smooth interior (Proposition ??). Let $\mu_{a,\beta,L}^k$ be the restricted Wilson measure on $\mathcal{M}_k^a$ and let $L_{a,\beta}^k$ be the associated Langevin generator.*

Then $\mu_{a,\beta,L}^k$ satisfies a logarithmic Sobolev inequality with constant

$$(9) \quad \rho_k(\beta, a, L) \geq (1 - \kappa_{\text{FP}}) m_0^2 - \delta(\beta, a, L),$$

where $\kappa_{\text{FP}} = 1/(2|\Phi^+(SU(N))|)$, $m_0^2 = (2\pi/L)^2$, and $\delta(\beta, a, L) \rightarrow 0$ as $\beta \rightarrow \infty$ and $a \rightarrow 0$ with L fixed. In the limit $L \rightarrow \infty$ with $a \rightarrow 0$ along the asymptotic-freedom trajectory, the limit measure on each non-trivial topological sector satisfies an LSI with constant $\rho_k^\infty \geq (1 - \kappa_{\text{FP}}) m_0^2(\infty) > 0$ provided $m_0^2(\infty) > 0$ persists in the limit (this is the transverse mass gap condition, weaker than the full Wilson mass gap, since it only requires positivity of the lowest eigenvalue on transverse harmonic 1-forms which is automatic on the lattice T^4 and on flat $\mathbb{R}^4$ in the appropriate Sobolev space).

Proof. By Theorem ??, the kernel of the Wilson Hessian (analogue of Lemma ?? in the lattice context, see Lüscher's gradient-flow construction of the lattice-instanton moduli space [?]) on the lattice instanton stratum $\mathcal{M}_k^a$ is exactly the tangent space to $\mathcal{M}_k^a$. The orthogonal complement Hessian has uniform lower bound $\lambda_\perp \geq (1 - \kappa_{\text{FP}}) m_0^2$ by Corollary ??.

By Proposition ??, the Ricci of $\mathcal{M}_k^a$ satisfies $\text{Ric} \geq (1 - \kappa_{\text{FP}}) m_0^2 g$ everywhere on the smooth interior.

By Theorem ??, the heat semigroup on $(\mathcal{M}_k^a, g_{\mathcal{M}_k^a})$ satisfies $\text{CD}((1 - \kappa_{\text{FP}}) m_0^2, \infty)$, and the heat-kernel measure has LSI with constant $(1 - \kappa_{\text{FP}}) m_0^2$.

By Proposition ??, the Wilson measure restricted to $\mathcal{M}_k^a$ is, up to exponentially small corrections at large β , the Riemannian volume measure on $\mathcal{M}_k^a$ weighted by the Hessian-Gaussian on the normal bundle, which is the heat-kernel measure at small flow time. The Holley–Stroock perturbation principle then transfers the LSI from the heat-kernel measure to the conditional Wilson measure, with a correction $\delta(\beta, a, L) \rightarrow 0$ encoding the perturbation. The continuum limit follows from the uniform-in- L nature of the Ricci bound and standard Fukushima–Oshima–Takeda descent. $\square$

Remark 5.2 (Comparison with conditional KR–FP–B). The constant in (??) is *identical in form* to the constant $(1 - \kappa_{\text{FP}}) m_0^2 / c_\infty(D)$ of [?, Thm. 6.1] on the trivial-sector projection, but is here *unconditional* because the AHS argument bypasses (H1) (generic vanishing) entirely on the instanton stratum. The $c_\infty(D)$ factor of the latter (which enters from the BBD LSI uniform descent) is replaced here by the direct Bakry–Émery descent on the finite-dimensional submanifold $\mathcal{M}_k$, which has no analogue of $c_\infty(D)$.

Corollary 5.3 (Explicit constant for $SU(3)$ on T^4). *For $G = SU(3)$ on the lattice torus T_L^4 at sufficiently large β and $k \in \mathbb{Z} \setminus \{0\}$,*

$$\rho_k(\beta, a, L) \geq \frac{5}{6} m_0^2 - \delta, \quad \kappa_{\text{FP}}(SU(3)) = 1/6.$$

6. THE $M = T^4$ CASE IN DETAIL

6.1. AHS on T^4 : caveat on H_A^2 . The original AHS argument [?] treats $M = S^4$ with $b_2^+(S^4) = 1$. For $M = T^4$ (the lattice background), $b_2^+(T^4) = 3$ and H_A^2 may be non-trivial; the index formula (??) requires the term $(1 + b_2^+)$ replaced by $(1 + b_2^+)$ generically, but the obstruction H_A^2 has to be checked.

For an *anti-self-dual* (ASD) connection on T^4 (by convention $F^- = 0$, replacing $F^+ = 0$ for the dual orientation), the cohomology H_A^2 governs obstructions in the deformation. For T^4 flat with the round product metric, generic flat $SU(N)$ -connections of charge $k = 0$ are smooth, but the higher-charge instantons must be approached carefully.

6.2. Periodic instantons (calorons) on T^4 . The natural generalisation of AHS to the 4-torus is the theory of periodic instantons (*calorons*); see Nahm [?] (citation to verify), van Baal [?] (citation to verify), Garcia-Perez–Gonzalez-Arroyo et al. [?] (citation to verify) for the lattice analogues. The Nahm transform provides a dual description identifying instantons on T^4 with solutions of Nahm’s equation on the dual torus $\hat{T}^4 = T^4$.

6.3. Status on T^4 : working hypothesis.

Hypothesis 6.1 (AHS rigidity on T^4). For $M = T^4$ at the lattice scale, the irreducible instanton moduli space $\mathcal{M}_k^a$ defined via the Lüscher gradient flow has, at every irreducible representative $[A]$, kernel of $\text{Hess } S_{YM}(A)$ exactly equal to $T_{[A]}\mathcal{M}_k^a$, with positive complement of size $\lambda_\perp \geq (1 - \kappa_{\text{FP}})m_0^2$, where $m_0^2 = (2\pi/L)^2$.

Remark 6.2 (Status of Hypothesis ??). On $M = S^4$ this is the original AHS theorem [?]. On $M = T^4$ this is *strongly expected* from the Nahm-transform duality (which identifies T^4 -instantons with Nahm-equation solutions on $\hat{T}^4$, preserving the irreducibility-rigidity structure) but the explicit verification of $H_A^2 = 0$ at all irreducible $[A] \in \mathcal{M}_k(T^4)$ to our knowledge is not a standard textbook result. The hypothesis is included to mark this as a transparent point requiring expert verification.

6.4. Conditional version of Theorem ?? on T^4 .

Theorem 6.3 (Conditional on Hypothesis ??). On $M = T^4$ with $G = SU(N)$ and the lattice Wilson measure as in (??), and under Hypothesis ??, the conclusion of Theorem ?? holds verbatim: $\mu_{a,\beta,L}^k$ satisfies LSI with constant $\rho_k(\beta, a, L) \geq (1 - \kappa_{\text{FP}})m_0^2 - \delta(\beta, a, L)$.

7. REMARKS AND OUTLOOK

7.1. Why this is *not* a full LSI for the Wilson measure. The full Wilson measure is supported on $\bigcup_{k \in \mathbb{Z}} \mathcal{M}_k^a$, and its decomposition gives $\mu_{a,\beta,L} = \sum_{k \in \mathbb{Z}} \pi_k \mu_{a,\beta,L}^k$ with $\pi_k = \mu_{a,\beta,L}(\mathcal{M}_k^a)$. Theorem ?? delivers an LSI on each $\mu_{a,\beta,L}^k$ for $k \neq 0$, but with constant ρ_k independent of π_k .

To deduce an LSI on the full mixture would require a Cheeger-type or Bakry–Ledoux mixing-inequality argument that combines the sector-wise LSIs with the discrete probability vector $(\pi_k)_k$. Such mixing-inequalities generally fail to give a useful uniform constant when one sector has π_k exponentially small relative to the other. In particular, the trivial sector $k = 0$ has $\pi_0 = 1 - O(e^{-8\pi^2/g^2})$ and dominates the measure, so its LSI constant controls the full Wilson LSI.

The *conclusion*: Theorem ?? does not produce an LSI on the full Wilson measure, but it provides a structurally clean unconditional sub-result that lives on $O(e^{-8\pi^2/g^2})$ mass.

7.2. Could AHS extend to the trivial sector $k = 0$? The natural attempt is to apply the deformation-complex argument to the trivial connection $A = 0$ on the trivial bundle. The cohomology $H_{A=0}^0 = \mathfrak{su}(N)$ (since $A = 0$ has full continuous isotropy: $\mathcal{G}$ -stabiliser is all global gauge transformations preserving $A = 0$, which is the constant subgroup $G \subset \mathcal{G}$, and its Lie algebra is $\mathfrak{su}(N)$ for G connected). So $A = 0$ is *not irreducible*, and the AHS hypothesis $H_A^0 = 0$ fails.

This is precisely the trivial-sector zero-mode obstruction discussed in [?, §5] and is the source of the hypothesis (H1) (generic vanishing). The AHS argument cannot bypass it without supplementary input.

7.3. Connection with the Cao–Park–Sheffield programme. The recent works of Cao–Park–Sheffield [?] and Cao–Nissim–Sheffield [?] construct scaling limits of $SU(N)$ lattice Yang–Mills via random-surface representations. Their construction is sector-wise transparent: the random-surface representation respects the topological charge decomposition (it sees the holonomies along closed loops, which distinguish topological sectors). Our Theorem ?? provides an unconditional LSI on each sector of their construction, which may be useful in their concentration estimates.

7.4. Connection with the Chatterjee Yang–Mills–Higgs result. Chatterjee [?] constructs a scaling limit of $SU(2)$ lattice Yang–Mills–Higgs in any $d \geq 2$, proving mass generation by the Higgs mechanism. The Higgs field breaks the symmetry that generates the AHS-trivial-sector zero mode, so the AHS argument adapts more cleanly to the Yang–Mills–Higgs measure: the trivial sector at $A = 0, \phi = \langle \phi \rangle$ has isotropy $\text{Stab}_{(A=0, \langle \phi \rangle)} = \text{Stab}_{\langle \phi \rangle} \subset G$, which is a *proper* subgroup of G . For the abelian Higgs with vacuum expectation $\langle \phi \rangle$ breaking G to a maximal torus, the residual isotropy is the Cartan, exactly the structure handled by the Kostant identity (Lemma KR–FP–2).

This suggests a follow-up project: *adapting the AHS argument to the (lattice) Yang–Mills–Higgs measure to obtain an unconditional LSI on the trivial Higgs vacuum sector*, with constant $\rho \geq (1 - \kappa_{\text{FP}}) m_0^2$ where κ_{FP} is now controlled by the *Higgs-broken* root system (not the full $\Phi^+(G)$). This would be a structural contribution to the Chatterjee scaling-limit programme.

7.5. Honest probability of conditional Clay closure. The result of this Letter, combined with the conditional reduction of [?], leaves the residual conditional Clay gap as follows:

- Trivial sector $k = 0$: requires (H1)–(H3) of [?] (one open hypothesis, two technical) + BBD-type uniform LSI on the lattice $SU(N)$ Wilson measure for the $k = 0$ stratum.
- Non-trivial sectors $k \neq 0$: *unconditional* via the present Theorem ??, modulo Hypothesis ?? on T^4 , which is expected to hold by Nahm-duality but requires expert verification.

The unconditional gain on $k \neq 0$ is of *negligible measure* ($O(e^{-8\pi^2/g^2})$ at large β), so the present result does *not* change the honest estimate $P(\text{Clay } 10\text{y}) \in [70\%, 82\%]$ post-Opus #319+ #2. Its value is mathematical, not statistical.

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